

#HW 2

● Graded

Student

Jacky Shen

 [View or edit group](#)

Total Points

94 / 100 pts

Question 1

(no title)

13 / 15 pts

– 0 pts Correct

– 2 pts There has been a mistake while you inter-converted the units. Your answer should be -4.63 Btu

– 0.5 pts Your final answer is not clear. You are supposed to write -4.63. The number 4 is not clear.

✓ – 2 pts You were supposed to convert your answer to Btu

– 7 pts The answer has not been submitted.

– 2 pts Answer should be in Btu.

Question 2

(no title)

15 / 15 pts

✓ – 0 pts Correct

– 7 pts Answer has not been submitted.

– 4 pts You have to find out the area of the plot. To calculate work you should multiply the volume difference with pressure. You have taken individual volumes and that should not be done.

– 2 pts You need to add up the areas. You have subtracted instead.

– 7 pts Answer has not been submitted.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts You have to calculate the work done in all the three steps and add them up. You have only done for the first step.
- 2 pts The work done will be negative in the first step. The total work should therefore be -2000 kJ.
- 2 pts It has been given $P(V^{1.5})$ constant. So $P = \text{constant}/(V^{1.5})$. You should apply that in integration of PdV in order to calculate work.
- 1 pt You should calculate the total work in all three steps.
- 2 pts There is an error in your calculation. The total work done will be -2000 kJ.
- 7 pts Answer has not been submitted.

Question 4

(no title)

11 / 15 pts

- 0 pts Correct

- 1 pt In part b E_2 will be 22 in order to get 15 as the difference.

✓ - 4 pts ΔE is the difference of two energies. E_1 is to be subtracted from E_2 . You should not add E_1 and E_2 .

- 1 pt in part e, E_1 should be +8 to get $\Delta E = 0$.

- 1 pt in c, ΔE should be -14 from $Q - W$, accordingly E_2 should be -8 to get ΔE as -14.

- 7 pts Answer has not been submitted.

Question 5

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 5 pts Answer has not been submitted.

- 1 pt You should calculate the total heat transfer in the process.

- 2 pts There is a mistake while you changed units. Answer should be -50 kJ.

Question 6

(no title)

30 / 30 pts

✓ - 0 pts Correct

- 15 pts No essay submitted

Question assigned to the following page: [6](#)

Jacky Shen

Doctor Bilgin

Chemical Engineering 201

September 17th

Role Model in Chemical Engineering

The Chemical Engineer role model I chose is Frances H. Arnold. Frances H. Arnold, born in 1956, is an American chemical engineer and Nobel prize winner. Frances is known for her work with “directed evolution of enzymes”, which is a method that creates proteins in the lab, while usually this process takes place in natural selection. The reason I find her inspiring is that I have always been a chemistry-focused student, while both of my parents work with the Biology side of science. Ever since I was a kid, my parents would talk about things like “lab-grown tissues” or “if science can figure that out, we can live much longer”. Frances can apply her Chemical Engineering knowledge to Biology, and to me, that is very inspiring.

Frances Hamilton Arnold was born in Pittsburgh, Pennsylvania, on July 25, 1956. She began her studies in mechanical & aerospace engineering at Princeton University in 1979, eventually shifting to chemical engineering, and earned her Ph.D. from UC Berkeley in 1983. After her doctorate, she started working at the California Institute of Technology in the mid-1980s. After joining the faculty at Caltech, Frances rose to become the Linus Pauling Professor of Chemical Engineering, Bioengineering, and Biochemistry. Linus Pauling was a very renowned chemist who won two Nobel Prizes in both Chemistry and Peace. Caltech named a professorship after him to honor his legacy, and being in this position meant that Caltech would permanently fund Frances’ research and teaching. It is one of the most honorable positions a faculty member can hold. Alongside this achievement, Frances also co-founded several biotech

Question assigned to the following page: [6](#)

companies, Gevo, Provivi, Aralez, to commercialize renewable fuels, green chemicals, and sustainable pest control. She also serves on the board of Alphabet, which is Google's parent company, as well as co-chair of the US President's Council of Advisors on Science and Technology. In 2018, she won the Nobel Prize in Chemistry. She was the first American Woman to receive this prize, for her work in directed evolution of enzymes.

Directed evolution is a laboratory technique that was developed and popularized by Frances. It allows scientists to improve or create enzymes by mimicking the process of natural selection, but much faster and more controllably. Enzymes are widely used in chemical engineering, pharmaceuticals, and industrial processes. It is a protein that speeds up chemical reactions. Natural enzymes don't normally perform well in an industrial environment, but directed evolution solves this problem by letting evolution itself search for better enzyme variants.

Frances H. Arnold – Facts – 2018. NobelPrize.org. Nobel Prize Outreach 2025. Thu. 18 Sep 2025. <<https://www.nobelprize.org/prizes/chemistry/2018/arnold/facts/>>

Questions assigned to the following page: [1](#), [2](#), and [3](#)

2.13.18 22 24 43 44

18

$$P_1 = 5 \text{ lbf/in}^2 \quad V_1 = 2.5 \text{ ft}^3 \quad P = 23.75 - 7.5V$$

$$P_2 = 20 \text{ lbf/in}^2 \quad V_2 = 0.5 \text{ ft}^3$$

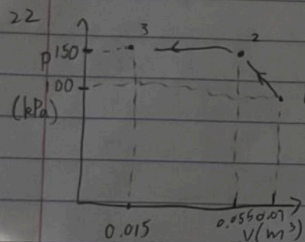
$$W = \int_{2.5}^{0.5} P dV = \int_{2.5}^{0.5} (23.75 - 7.5V) dV = \left(23.75V - \frac{7.5V^2}{2} \right) \Big|_{2.5}^{0.5}$$

$$= \left(23.75 \cdot 0.5 - \frac{7.5(0.5)^2}{2} \right) - \left(23.75 \cdot 2.5 - \frac{7.5(2.5)^2}{2} \right)$$

$$= (11.875 - 0.9375) - (59.375 - 23.4375)$$

$$= 10.9375 - 35.9375$$

$$= -25 \text{ psi} \cdot \text{ft}^3$$



$$P_1 = 100 \quad V_1 = 0.07$$

$$P_2 = 150 \quad V_2 = 0.055$$

$$P_3 = 150 \quad V_3 = 0.015$$

$$(150)(0.015 - 0.055) + (100)(0.055 - 0.07) + (150 - 100)(0.055 - 0.07) \left(\frac{1}{2} \right)$$

$$= -6 + -1.5 + -0.375 = -7.875 \text{ kPa} \cdot \text{m}^3 = \boxed{-7.875 \text{ kJ}}$$

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Process 1-2 $P_1 = 10 \text{ bar} \rightarrow P V^{1.5} = \text{constant} \quad P = \frac{\text{constant}}{V^{1.5}}$

$$V_1 = 1 \text{ m}^3 \quad V_2 = 4 \text{ m}^3 \quad P_2 = 10$$

$$W = \int_1^4 \frac{\text{constant}}{V^{1.5}} dV = 10 \cdot \int_1^4 V^{-1.5} dV \quad (0.1^{1.5} = 10 \quad P_2 = 1.25)$$

$$10 \left(-2V^{-0.5} \right) \Big|_1^4 = 10 \left((-2(4)^{-0.5}) - (-2(1)^{-0.5}) \right) = 10(-1 + 2) = 10 \text{ bar} \cdot \text{m}^3$$

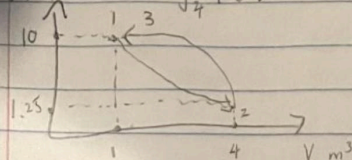
$$= \boxed{1000 \text{ kJ}}$$

Process 2-3 $V_2 = V_3, \quad W = 0, \quad P_3 = 10 \text{ bar}$

Process 3-1 $P_3 = 10 \text{ bar} \rightarrow P_1 = 10 \text{ bar}$

$$V_3 = V_2 = 4 \text{ m}^3 \rightarrow V_1 = 1 \text{ m}^3$$

$$W = \int_3^1 P dV = 10 \int_4^1 dV = 10(1 - 4) = -30 \text{ bar} \cdot \text{m}^3 = \boxed{-3000 \text{ kJ}}$$



$$1000 \text{ kJ} + -3000 + 0 = \boxed{-2000 \text{ kJ}}$$

Questions assigned to the following page: [4](#) and [5](#)

43	Q	W	E ₁	E ₂	ΔE	a
a	20	-25	15	30	45	$\Delta E = E_1 + E_2 = Q - W$
b	-27	-32	7	8	15	$\Delta E = 15 + 30 = Q - (-25) = 45$
c	-4	10	6	-20	-14	$Q + 25 = 45 \quad Q = 20$
d	10	-10	10	10	20	$15 - 7 = 8$
e	3	3	-8	8	0	$-27 - W = 15$
						$-W = 32$
						$c \begin{pmatrix} -4 - 10 = -14 = \Delta E \\ -14 - 6 = -20 \end{pmatrix}$
						$d \begin{pmatrix} 20 - 10 = 10 \\ Q + 10 = 20 \quad Q = 10 \end{pmatrix}$
						$e \begin{pmatrix} 0 - 8 = -8 \quad 3 - W = 0 \quad W = 3 \end{pmatrix}$

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10 kg work by system, +0.147 kJ/kg
elevation -50m, V 15m/s $\rightarrow$ 30m/s V decreases 5kJ/kg
 $g = 9.7 \text{ m/s}^2$ heat transfer?

$$Q - W = \Delta E = \Delta KE + \Delta U + \Delta PE$$

$$W = 10 \text{ kg} \cdot 0.147 \text{ kJ/kg} = 1.47 \text{ kJ}$$

$$\Delta U = -5 \text{ kJ/kg} \cdot 10 = -50 \text{ kJ}$$

$$\Delta PE = 9.7 \cdot (-50) = -485 \text{ J/kg} = -0.485 \text{ kJ/kg} \quad -0.485 \cdot 10 = -4.85 \text{ kJ}$$

$$\Delta KE = \frac{1}{2} (30^2 - 15^2) = 337.5 \text{ J/kg} = 0.3375 \text{ kJ/kg} \quad 0.3375 \cdot 10 = 3.375 \text{ kJ}$$

$$Q - 1.47 = 3.375 - 50 - 4.85$$

$$Q = -50.005 \text{ kJ}$$